

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0654

Saltery Bay 138 kV
July 21, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0654. The Interconnection Customer proposes a 894.7 MW natural gas combined cycle facility interconnecting at the Saltery Bay 138 kV substation in the West Virginia area. The request was received on July 27, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 218 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$76,292,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Phelan – Gauley Mtn 138kV	Saltery Bay – Hayfork 138kV	345.5	204	73.0
Saltery Bay – Gauley Mountain 138kV	Hayfork – Saltery Bay 138kV	337.2	210	73.0
Saltery Bay – Hayfork 138kV	Saltery Bay – Gauley Mountain 138kV	284.4	242	74.2
Hayfork Substation – Hawks Nest Hydro 138kV	Saltery Bay – Gauley Mtn Substation 138kV Line	278.1	246	74.2
Phelan – Bradley Substation 138kV	Scarbrough – N Sweetwater Substation 138kV L	166.1	230	12.9
Alpha Rd – Pawnee 138kV	Beech Ridge Energy Llc 138/1kV TX	162.9	177	5.4
Bradley – American Falls 138kV	N Sweetwater – Scarbro 138kV	137.4	179	9.2
Mattoon East – Lexington Health Center 138kV	Amos – Coulee Ave 345kV L	132.3	216	5.5
Heyburn – Lysander 138kV line	Coburn – Cabin Creek 345kV	132.0	191	5.6
Dixieland Prison – Lexington Health Ctr 138kV	Coulee Ave – Amos 345kV Line	130.8	221	5.5
Big Basin – Sherman Rd 138kV	Donnells-Curtis – Beech Ridge Energy Llc 138kV	130.3	243	5.3
Chemical – Alum Creek 138kV	Brownfield Switching – Eagle Vly In 138kV	129.8	316	5.2
New Guf – Dixieland Prison 138kV	Coulee Ave – Amos 345kV line	129.7	225	5.5
Beech Ridge Energy Llc – Big Basin 138kV	Donnells-Curtis – Beech Ridge Energy Llc 138kV	129.6	246	5.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Congress St – Lake Hemet Substation 138kV	Beech Ridge Energy Llc 138/1kV Transformer Bank	125.9	232	7.2
Congress St – Pawnee Substation 138kV	Beech Ridge Energy Llc 138-1 kV TX	125.5	231	7.2
Lake Hemet – Reedy Crk 138kV	Beech Ridge Energy Llc 138/1 kV XFMR	125.4	234	7.2
Vergennes 9 – Reedy Creek 138kV	Beech Ridge Energy Llc 138-1 kV XFMR	125.1	236	7.2
Beech Ridge Energy Llc 345/1kV Transformer Bank	Vergennes 9 345-138 kV Transformer	123.7	397	5.1
Raver – Lysander 138kV Line	Coburn – Cabin Crk 345kV line	120.0	217	5.6
International Falls Power – Laurelwood 138kV line	Brownfield Switching – Eagle Valley In 138kV	119.8	208	9.7
Raver – Sherman Rd Substation 138kV	Cabin Creek – Cobun 345kV	118.7	222	5.6
Malad – Dorris 138kV	Coulee Ave – Amos 345kV	117.8	269	6.7
Beech Ridge Energy Llc 138/1kV Transformer Bank	Vergennes 9 345/138 kV TX	115.7	425	5.1
Stone Corral – Shingle Springs 138kV	Saltery Bay – Gauley Mtn 138kV line	115.2	371	17.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Phelan – Gauley Mtn 138kV Line	Rebuild line with higher-capacity conductor	\$7,862,000	94% (\$7,411,000)
Gauley Mountain – Saltery Bay 138kV	Rebuild line with higher-capacity conductor	\$5,005,000	94% (\$4,718,000)
Saltery Bay – Hayfork 138kV	Rebuild line with higher-capacity conductor	\$6,345,000	92% (\$5,826,000)
Hayfork – Hawksnest Hydro 138kV	Rebuild line with higher-capacity conductor	\$4,522,000	92% (\$4,152,000)
Phelan – Bradley 138kV	Rebuild line with higher-capacity conductor	\$21,751,000	63% (\$13,646,000)
Alpha Rd – Pawnee 138kV	Rebuild line with higher-capacity conductor	\$4,359,000	35% (\$1,525,000)
Bradley – American Falls 138kV	Rebuild line with higher-capacity conductor	\$5,456,000	100% (\$5,456,000)
Lexington Health Ctr – Mattoon East 138kV	Rebuild line with higher-capacity conductor	\$11,701,000	22% (\$2,612,000)
Heyburn – Lysander 138kV	Rebuild line with higher-capacity conductor	\$12,376,000	30% (\$3,703,000)
Dixieland Prion – Lexington Health Center 138kV	Rebuild line with higher-capacity conductor	\$10,865,000	22% (\$2,426,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Big Basin – Sherman Rd 138kV	Rebuild line with higher-capacity conductor	\$9,092,000	27% (\$2,420,000)
Alum Crk – Chemical 138kV	Rebuild line with higher-capacity conductor	\$9,931,000	9% (\$905,000)
New Gulf – Dixieland Prison 138kV L	Rebuild line with higher-capacity conductor	\$5,021,000	22% (\$1,121,000)
Beech Ridge Energy Llc – Big Basin 138kV	Rebuild line with higher-capacity conductor	\$11,690,000	27% (\$3,112,000)
Lae Hemet – Congress St 138kV	Rebuild line with higher-capacity conductor	\$3,182,000	34% (\$1,089,000)
Congress St – Pawnee Substation 138kV	Rebuild line with higher-capacity conductor	\$5,156,000	34% (\$1,765,000)
Reedy Crk – Lake Hemet 138kV	Rebuild line with higher-capacity conductor	\$8,136,000	34% (\$2,785,000)
Reedy Crk Substation – Vergennes 9 138kV	Rebuild line with higher-capacity conductor	\$4,602,000	34% (\$1,575,000)
Beech Ridge Energy Llc 345/1kV Transformer Bank	Replace transformer with higher-rated unit	\$14,511,000	16% (\$2,303,000)
Lysander – Raver 138kV	Rebuild line with higher-capacity conductor	\$2,865,000	30% (\$857,000)
Laurelwood – International Falls Power 138kV Line	Rebuild line with higher-capacity conductor	\$3,819,000	26% (\$982,000)
Raver – Sherman Rd Substation 138kV	Rebuild line with higher-capacity conductor	\$3,839,000	30% (\$1,149,000)
Malad – Dorris 138kV line	Rebuild line with higher-capacity conductor	\$2,666,000	20% (\$543,000)
Beech Ridge Energy Llc 138/1kV Transformer Bank	Replace transformer with higher-rated unit	\$7,554,000	16% (\$1,199,000)
Shingle Springs – Stone Corral 138kV	Rebuild line with higher-capacity conductor	\$3,012,000	100% (\$3,012,000)
	Total allocated to project		\$76,292,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.